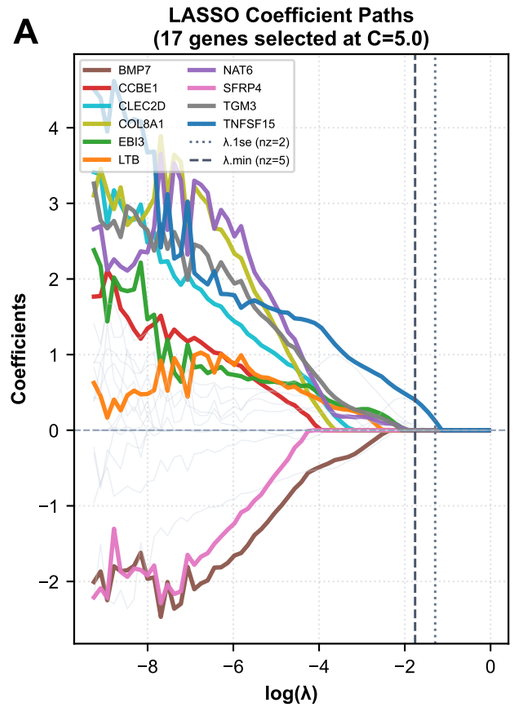
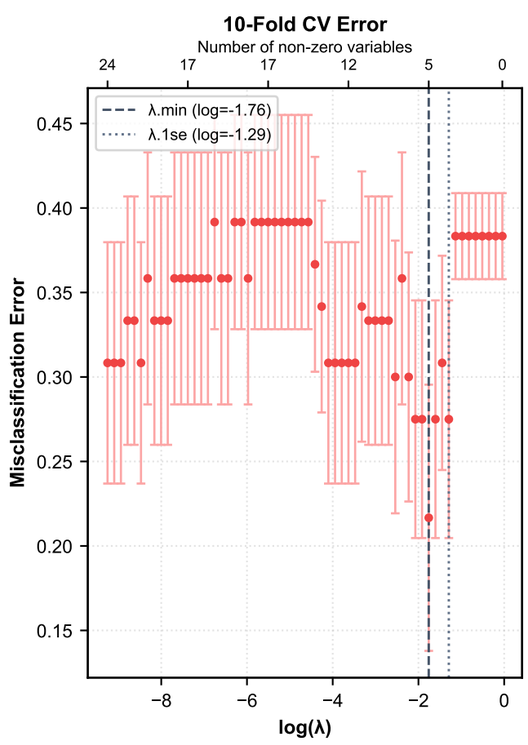
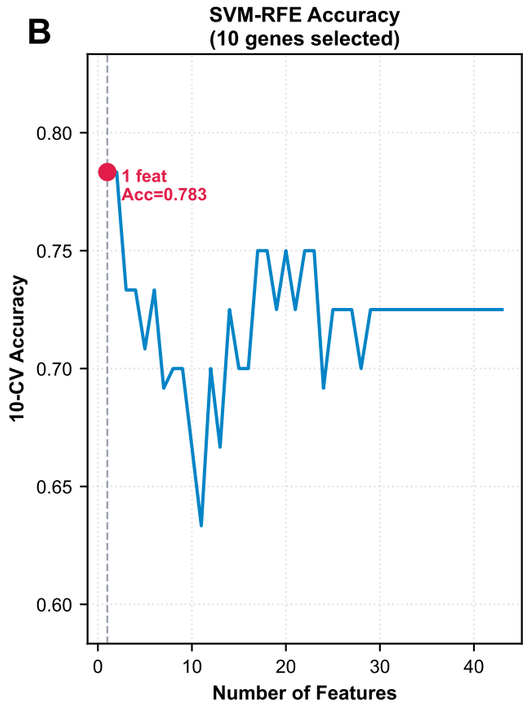
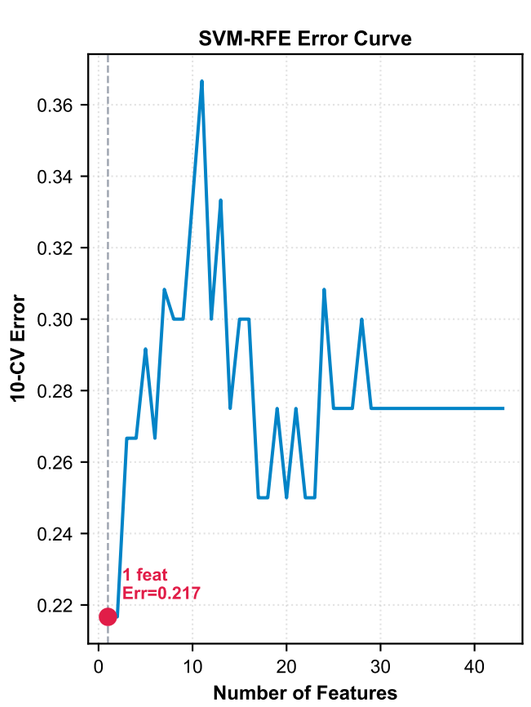
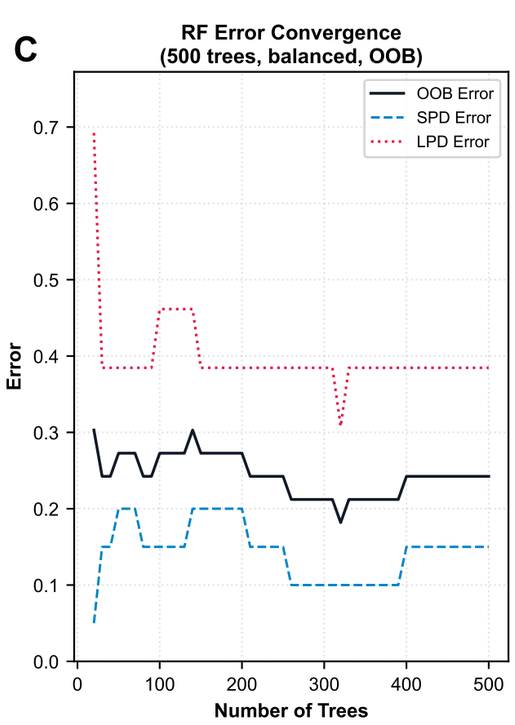
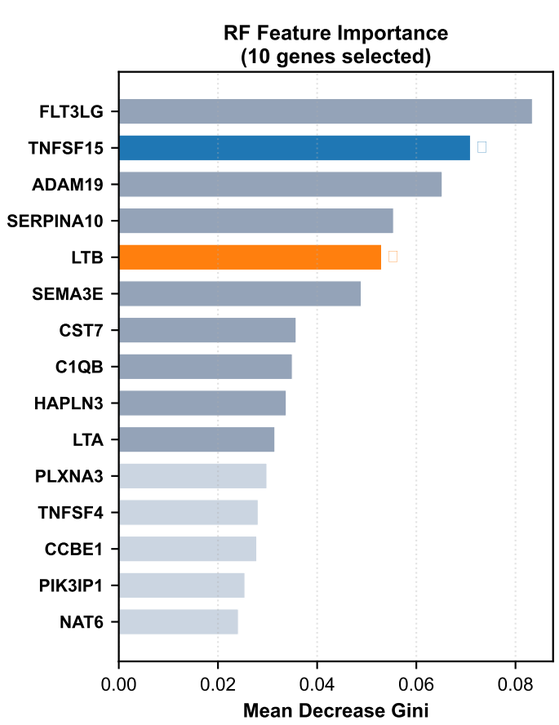

A
LASSO Coefficient Paths
(17 genes selected at C=5.0)
BMP7
CCBE1
CLEC2D
COL8A1
EBI3
LTB
NAT6
SFRP4
TGM3
TNFSF15
λ.1se (nz=2)
λ.min (nz=5)
Coefficients
log(λ)

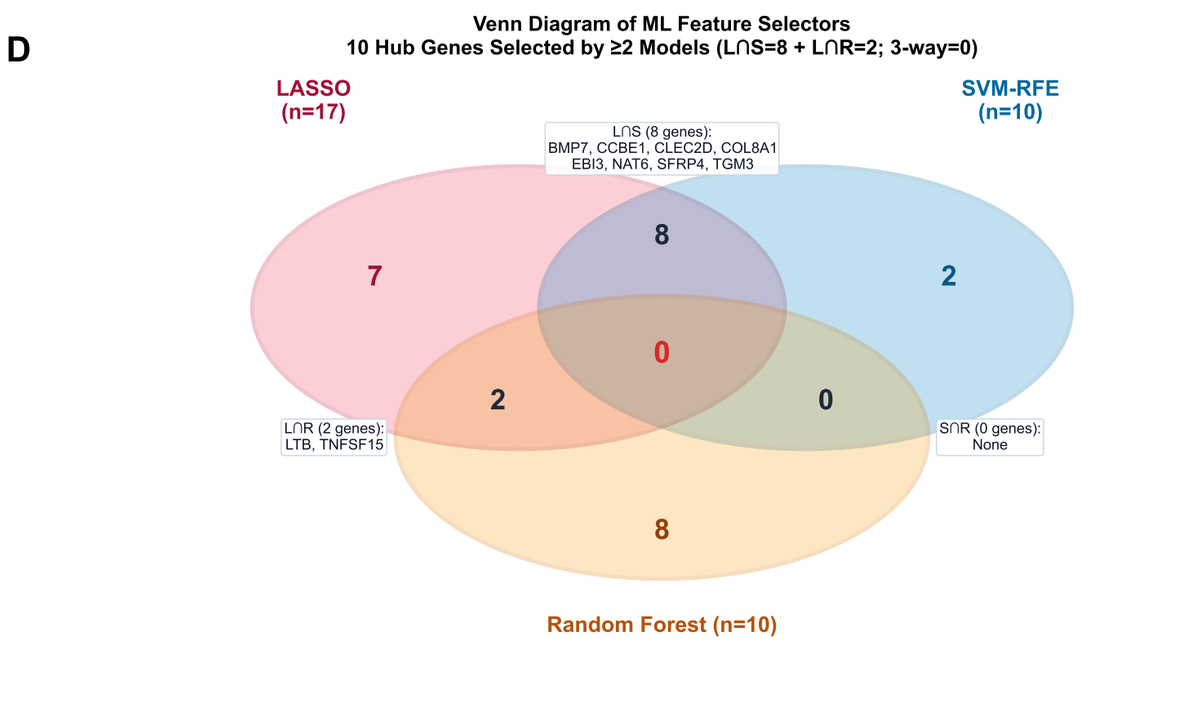

10-Fold CV Error
Number of non-zero variables
24 17 17 12 5 0
λ.min (log=-1.76)
λ.1se (log=-1.29)
Misclassification Error
log(λ)

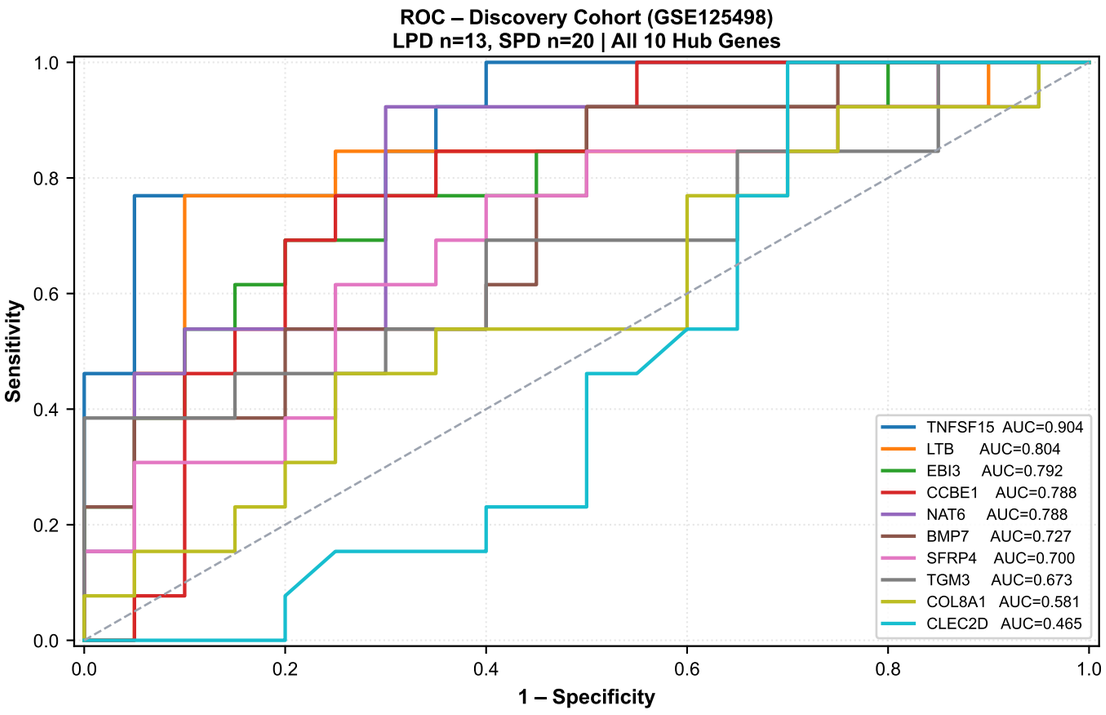

B
SVM-RFE Accuracy
(10 genes selected)
1 feat
Acc=0.783
10-CV Accuracy
Number of Features

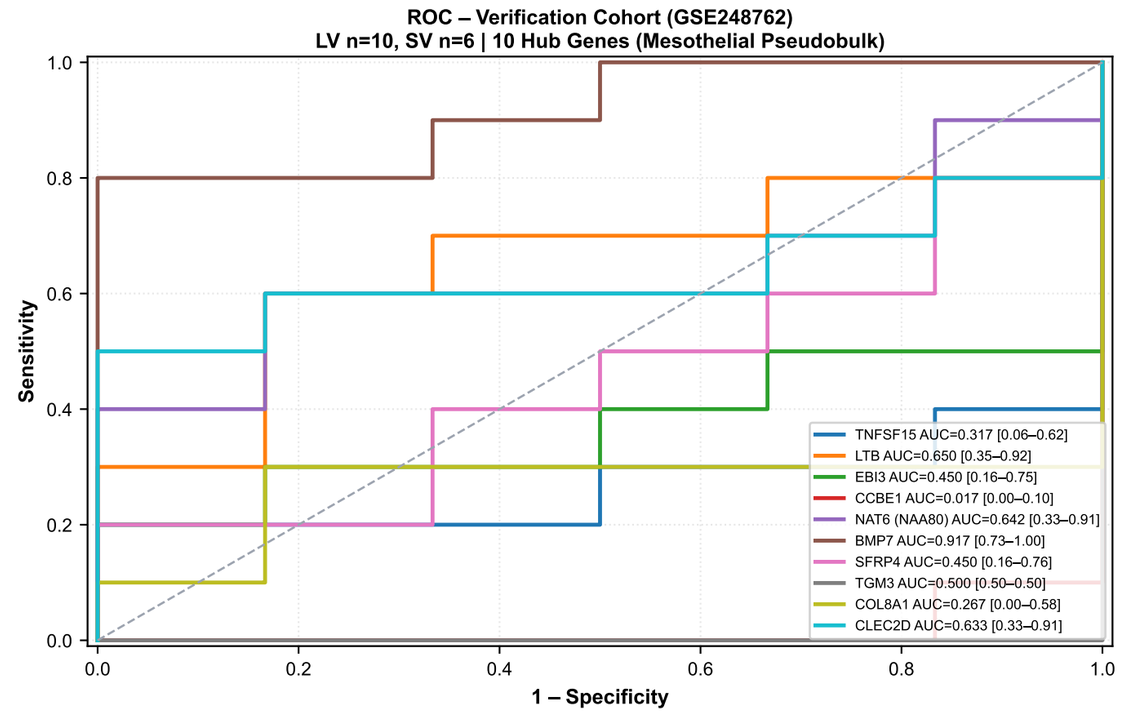

SVM-RFE Error Curve
1 feat
Err=0.217
10-CV Error
Number of Features

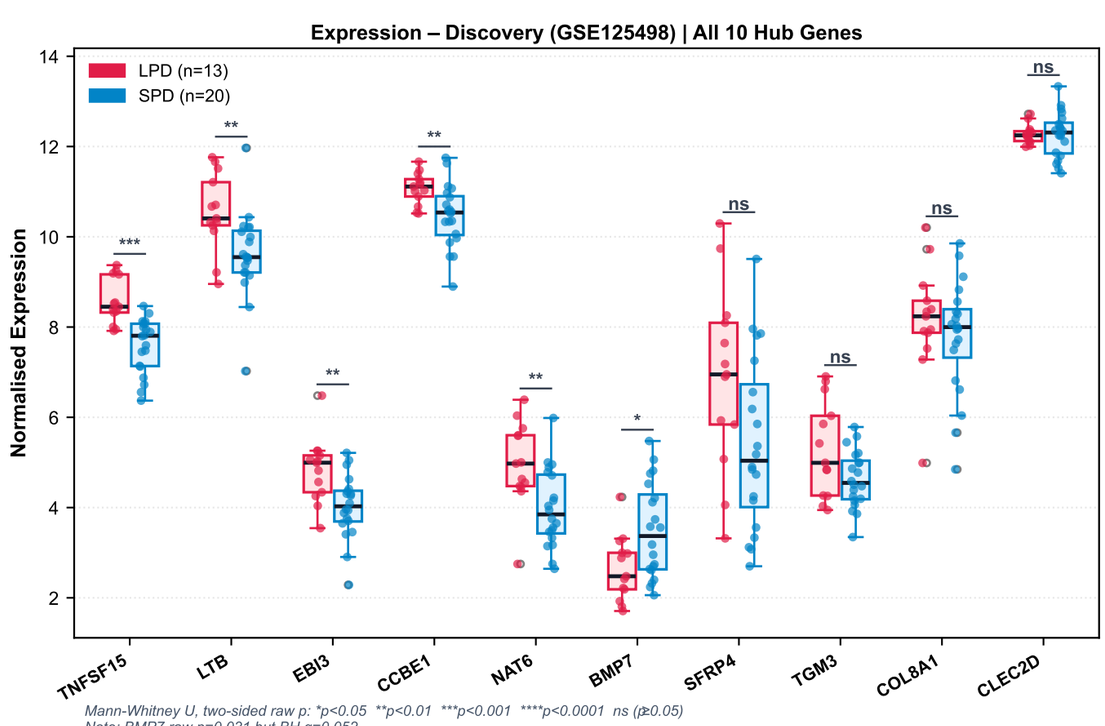

C
RF Error Convergence
(500 trees, balanced, OOB)
OOB Error
SPD Error
LPD Error
Error
Number of Trees

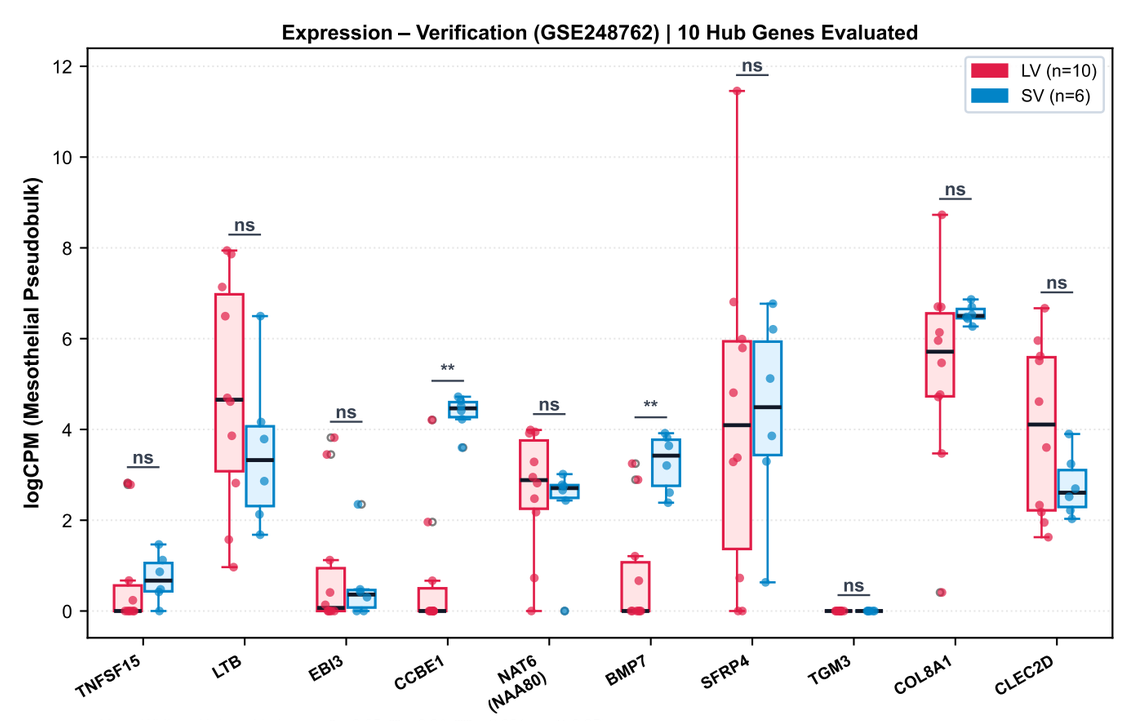

RF Feature Importance
(10 genes selected)
FLT3LG
TNFSF15
ADAM19
SERPINA10
LTB
SEMA3E
CST7
C1QB
HAPLN3
LTA
PLXNA3
TNFSF4
CCBE1
PIK3IP1
NAT6
Mean Decrease Gini


D
Venn Diagram of ML Feature Selectors
10 Hub Genes Selected by ≥2 Models (L∩S=8 + L∩R=2; 3-way=0)
LASSO
(n=17)
SVM-RFE
(n=10)
L∩S (8 genes):
BMP7, CCBE1, CLEC2D, COL8A1
EBI3, NAT6, SFRP4, TGM3
7
8
2
0
2
0
L∩R (2 genes):
LTB, TNFSF15
S∩R (0 genes):
None
8
Random Forest (n=10)


E
ROC – Discovery Cohort (GSE125498)
LPD n=13, SPD n=20 | All 10 Hub Genes
TNFSF15 AUC=0.904
LTB AUC=0.804
EBI3 AUC=0.792
CCBE1 AUC=0.788
NAT6 AUC=0.788
BMP7 AUC=0.727
SFRP4 AUC=0.700
TGM3 AUC=0.673
COL8A1 AUC=0.581
CLEC2D AUC=0.465
Sensitivity
1 – Specificity


F
ROC – Verification Cohort (GSE248762)
LV n=10, SV n=6 | 10 Hub Genes (Mesothelial Pseudobulk)
TNFSF15 AUC=0.317 [0.06–0.62]
LTB AUC=0.650 [0.35–0.92]
EBI3 AUC=0.450 [0.16–0.75]
CCBE1 AUC=0.017 [0.00–0.10]
NAT6 (NAA80) AUC=0.642 [0.33–0.91]
BMP7 AUC=0.917 [0.73–1.00]
SFRP4 AUC=0.450 [0.16–0.76]
TGM3 AUC=0.500 [0.50–0.50]
COL8A1 AUC=0.267 [0.00–0.58]
CLEC2D AUC=0.633 [0.33–0.91]
Sensitivity
1 – Specificity


G
Expression – Discovery (GSE125498) | All 10 Hub Genes
LPD (n=13)
SPD (n=20)
Normalised Expression
TNFSF15
LTB
EBI3
CCBE1
NAT6
BMP7
SFRP4
TGM3
COL8A1
CLEC2D
Mann-Whitney U, two-sided raw p: *p<0.05 **p<0.01 ***p<0.001 ****p<0.0001 ns (p≥0.05)
Note: BMP7 raw p=0.031 but BH q=0.052


H
Expression – Verification (GSE248762) | 10 Hub Genes Evaluated
LV (n=10)
SV (n=6)
logCPM (Mesothelial Pseudobulk)
TNFSF15
LTB
EBI3
CCBE1
NAT6 (NAA80)
BMP7
SFRP4
TGM3
COL8A1
CLEC2D
Mann-Whitney U, two-sided raw p: *p<0.05 **p<0.01 ***p<0.001 ns (p≥0.05)